**COURSE STRUCTURE BACHELOR in SOFTWARE ENGINEERING with HONOURS**
**4 YEAR PROGRAMME, INTAKE SEPTEMBER 2024/2025**

| **YEAR 1: SEMESTER 1*** ** | | | |
|---|---|---|---|
| **Code** | **Course** | **Credit** | **Pre-requisite** |
| SCSE1013 | Fundamental Programming Concept | 3 | |
| SCST1123 | Mathematics for Software Engineer | 3 | |
| SCSR1013 | Digital Logic | 3 | |
| SCST1143 | Database Engineering | 3 | |
| ULRS1032 | Integrity and Anti-Corruption | 2 | |
| Sxxxxxx3 | Free Elective I* | 3 | |
| | **TOTAL CREDIT** | **17** | |
| | **CUMULATIVE CREDITS** | **17** | |

****Students with IELTS Band less than 5.5 or TOEFL less than 525 or TOEFL IBT less than 60 or CEFR less than B2 or MUET less than Band 4 {Malaysian student) must register for UHLB1112 course*

** Students must choose University Free Electives subjects offered by faculties other than Faculty of Computing.*

| **YEAR 1: SEMESTER 2** | | | |
|---|---|---|---|
| **Code** | **Course** | **Credit** | **Pre-requisite** |
| SCSE1203 | Software Engineering Principles | 3 | SCSE 1013 |
| SCSR1033 | Computer Organization and Architecture | 3 | SCSR1013 |
| SCST1223 | Probability and Statistical Data Analysis | 3 | |
| SCSE1224 | Advanced Programming | 4 | SCSE1013 |
| SCSR2213 | Network Communications | 3 | |
| ***Malaysian Students*** | | | |
| ULRS1182 | Appreciation of Ethics and Civilizations | 2 | |
| ***International Students*** | | | |
| UHLM1012 | Malaysia Language for Communication 2 | 2 | |
| | **TOTAL CREDIT** | **18** | |
| | **CUMULATIVE CREDITS** | **35** | |

| YEAR 2: SEMESTER 1 | | | |
|---|---|---|---|
| **Code** | **Course** | **Credit** | **Pre-requisite** |
| SCSE2133 | Software Process and Project Management | 3 | SCSE 1203 |
| SCSE2123 | Software Requirements Engineering | 3 | SCSE 1203 |
| SCSE2103 | Data Structure and Algorithm | 3 | SCSE1013 |
| SCSR2043 | Operating Systems | 3 | |
| SCSM2113 | Human Computer Interaction Fundamentals | 3 | SCSE 1203 |
| ULRF2xx2 | Service Learning & Community Engagement Courses | 2 | |
| | **TOTAL CREDIT** | **17** | |
| | **CUMULATIVE CREDITS** | **52** | |

| YEAR 2: SEMESTER 2 | | | |
|---|---|---|---|
| **Code** | **Course** | **Credit** | **Pre-requisite** |
| SCSM2223 | Cross-Platform Application Development | 3 | SCST 1143 |
| SCSE2233 | Software Design & Architecture | 3 | SCSE1203 |
| SCSE2243 | Application Development Project I | 3 | SCSE1203, SCSE2123 |
| UHLB2122 | Professional Communication Skills1 | 2 | |
| ULRS1022 | Philosophy and Current Issues | 2 | |
| **Program Electives – Choose 1 (3 credits)** | | | |
| SCSB2103 | Bioinformatics I | 3 | |
| SCSP2753 | Data Mining | 3 | |
| SCSP3213 | Business Intelligence | 3 | |
| | **TOTAL CREDIT** | **16** | |
| | **CUMULATIVE CREDITS** | **68** | |

| YEAR 3: SEMESTER 1 | | | |
|---|---|---|---|
| **Code** | **Course** | **Credit** | **Pre-requisite** |
| UHLB3132 | Professional Communication Skills | 2 | |
| UHLx1122 | Foreign Language Elective | 2 | |
| Sxxxxxx3 | Free Elective II* | 3 | |
| **Program Electives – Choose 3 (9 credits)** | | | |
| SCST3223 | Data Analytic Programming | 3 | |
| SCSE3143 | Ubiquitous Computing | 3 | |
| SCSR3113 | Cloud Computing | 3 | |
| SCSE3103 | Cognitive Computing | 3 | |
| SCSE3203 | Special Topics | 3 | |
| SCSM3113 | Virtual and Augmented Reality Application | 3 | |
| | **TOTAL CREDIT** | **16** | |
| | **CUMULATIVE CREDITS** | **84** | |
| ** Students must choose University Free Electives subjects offered by faculties other than Faculty of Computing.* | | | |

| YEAR 3: SEMESTER 2 | | | |
|---|---|---|---|
| **Code** | **Course** | **Credit** | **Pre-requisite** |
| SCSE3242 | Software Engineering Project I | 2 | 80 credits<br>SCSE2243 |
| SCSR3133 | Secure Software Programming | 3 | SCSM 2213 |
| SCSE3213 | Software Quality & Testing | 3 | SCSE 2123<br>SCSE 2233 |
| SCSE3223 | Application Development Project II | 3 | SCSE2243 |
| SCSE3233 | Professional Practice in Software Engineering | 3 | SCSE 1203 |
| ULRS3032 | Entrepreneurship and Innovation | 2 | |
| | **TOTAL CREDIT** | **16** | |
| | **CUMULATIVE CREDITS** | **100** | |

| YEAR 4: SEMESTER 1 | | | |
|---|---|---|---|
| **Code** | **Course** | **Credit** | **Pre-requisite** |
| SCSE4108 | Industrial Training (HW) | 8 | 92 credits CGPA >= 2.0 |
| SCSE4114 | Industrial Training Report | 4 | 92 credits CGPA >= 2.0 |
| | **TOTAL CREDIT** | **12** | |
| | **CUMULATIVE CREDITS** | **112** | |

| YEAR 4: SEMESTER 2 | | | |
|---|---|---|---|
| **Code** | **Course** | **Credit** | **Pre-requisite** |
| SCSE4214 | Software Engineering Project II | 4 | SCSE3242 |
| Sxxxxxx3 | Free Elective III | 3 | |
| Sxxxxxx3 | Free Elective IV | 3 | |
| **Program Electives – Choose 2 (6 credits)** | | | |
| SCSR4453 | Network Security | 3 | |
| SCSR 4973 | Computer Network & Security Special Topics | 3 | |
| SECB3133 | Computational Biology I | 3 | |
| SECB3203 | Programming for Bioinformatics | 3 | |
| SECR5xx3/ SECP5xx3/ SECJ5xx3 | PRISMS Elective 1 | 3 | |
| SECR5xx3/ SECP5xx3/ SECJ5xx3 | PRISMS Elective 2 | 3 | |
| | **TOTAL CREDIT** | **16** | |
| | **CUMULATIVE CREDITS** | **128** | |

**PRISMS Elective Courses**

The 5000-level courses indicate elective courses that are equivalent to core courses at the master's level. Please refer to the handout on UTM's Bachelor-Master Integration Program (PRISMS) for further details, including eligibility. Please also consult your academic advisor for better clarification.

i. SECR5033 Information Security Governance and Risk Management
ii. SECR5043 Cloud Computing Security
iii. SECJ5013 Secure Software Engineering
iv. SECR5053 Penetration Testing
v. SECJ5023 Advanced Theory of Computer Science
vi. SECJ5033 Advanced Data Structure and Algorithms
vii. SECJ5043 Advanced Artificial Intelligence
viii. SECP5013 Advanced Analytics for Data Science
ix. SECP5023 Big Data Management
x. SECP5033 Business Intelligence and Analytics
xi. SECP5043 Data Science Governance
xii. SECP5053 Massive Mining and Streaming
xiii. SECP5063 Statistics for Data Science